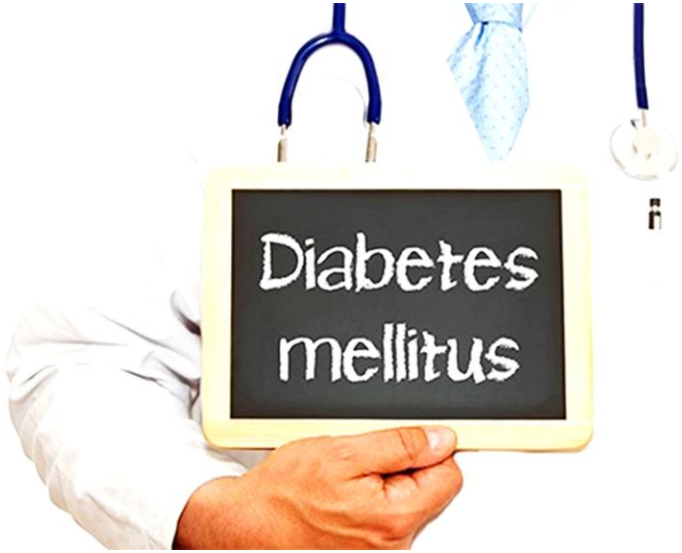
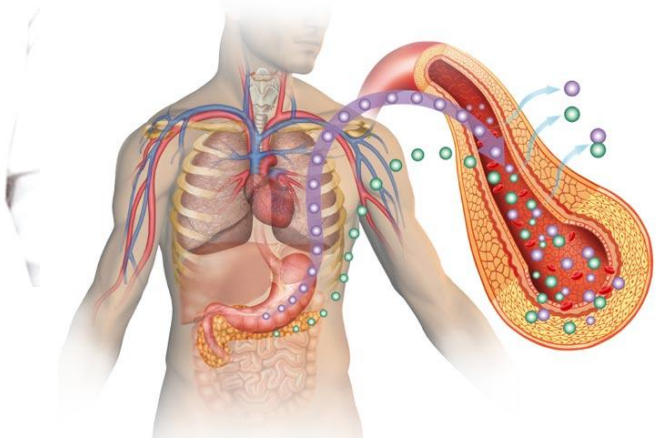
Laboratory Investigations

in the Diagnosis and Management of Diabetes





Assoc. Prof. Dr. Wilaiwan Sriwimol

Department of Pathology, Faculty of Medicine

Prince of Songkla University

(sriwimol.wi@gmail.com)

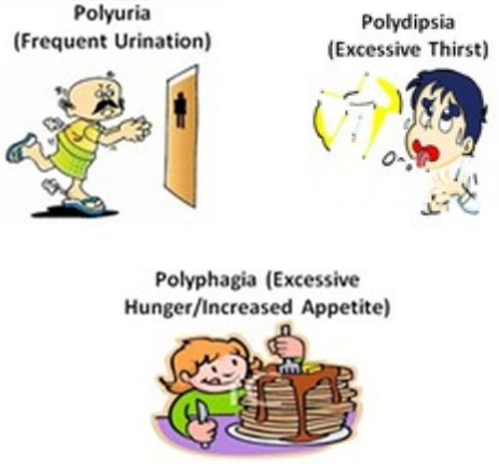
- **Lecture 10**: Hormones of pancreatic islets (อ.พิศเรศ)
- **SDL 2**: Blood glucose control and balance (อ.ฐณะวัฒน์)
- **SDL 3**: Diabetes mellitus (DM) (อ.สุภมัย)

SDL 4: Laboratory investigations in the diagnosis and management of diabetes

วัตถุประสงค์:

อธิบายหลักการทดสอบ เลือกใช้และแปลผลการทดสอบทางห้องปฏิบัติการเพื่อวินิจฉัยและติดตามดูแลผู้ป่วยเบาหวานได้

Diabetes Mellitus (DM)



Diabetes mellitus (DM) is a group of metabolic disorder sharing the common features of hyperglycemia.

- Impaired insulin secretion
- Impaired insulin function

❑ Type 1 DM

Beta-cell destruction, usually leading to absolute insulin deficiency

❑ Type 2 DM

Predominantly insulin resistance with relative insulin deficiency to a predominantly secretion

❑ Other specific types

- Genetic defects of beta-cell function
- Genetic defects in insulin action
- Diseases of the exocrine pancreas
- Endocrinopathies
- Drug or chemical induced
- Infections
- Uncommon forms of immune mediated diabetes
- Other genetic syndromes sometimes associated with diabetes

❑ Gestational diabetes

Laboratory Tests for DM

- Diagnosis of DM
- Screening of DM
- Assessment of glycemic control
- Assessment of long-term risks
- Assessment of acute metabolic complications

Blood

- Plasma glucose: random, fasting, 2-h postprandial
- Glucose tolerance test (GTT)
- HbA_1c
- Fructosamine
- C-peptide

etc.

Urine

- Glucose
- Microalbumin
- Ketone

etc.

Criteria for Diagnosis of DM

1. Typical symptoms of DM (polyuria, polydipsia, weight loss) and random plasma glucose ≥ 200 mg/dL (≥ 11.1 mmol/L)
2. Fasting plasma glucose ≥ 126 mg/dL (≥ 7.0 mmol/L)
3. 2-hour post glucose load (75 g) value during oral glucose tolerance test ≥ 200 mg/dL (≥ 11.1 mmol/L).

If any one of the above three criteria is present, confirmation by repeat testing on a subsequent day is necessary for establishing the diagnosis of DM.

4. HbA1c ≥ 6.5%*

Laboratory Tests for Diagnosis of DM

- Estimation of blood glucose
- Oral glucose tolerance test (OGTT)

Estimation of Blood Glucose

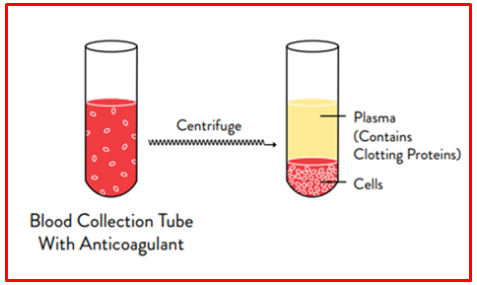
- Measurement of blood glucose is indicative of *current state* of carbohydrate metabolism.

Whole blood (from venous blood)



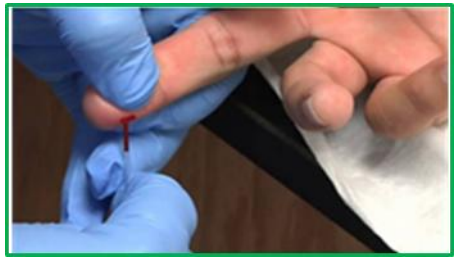
- Affected by protein concentration (especially hemoglobin)
- Affected by hematocrit

Plasma (from venous blood)



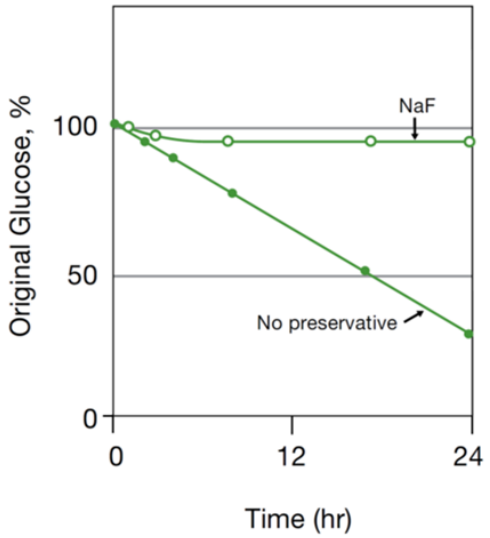
Plasma is preferred for estimation of glucose

Capillary blood

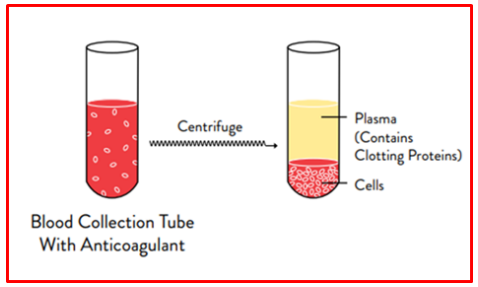


Plasma Glucose


NaF tube



>> To collect venous blood into the NaF tube



- When whole blood is left at room temperature after collection, glycolysis reduces glucose level at the rate of about 7 mg/dL/h.
- Glycolysis is further increased in the presence of bacterial contamination or leukocytosis.

>> To rapid separate the plasma from cell

>> To rapid measure the plasma glucose concentration

Plasma Glucose

Test	Time of blood collection	Interpretation
Fasting plasma glucose (FPG)	After 8-h fast (after an overnight fast)	Normal: 70 – 99 mg/dL (<100 mg/dL) Impaired fasting glucose: 100 – 125 mg/dL Diabetic mellitus: ≥ 126 mg/dL
Random blood sugar (random blood glucose, causal plasma glucose)	Any time of the day	Normal: < 140 mg/dL Abnormal: ≥ 200 mg/dL

Methods for Plasma Glucose Measurement

Chemical method

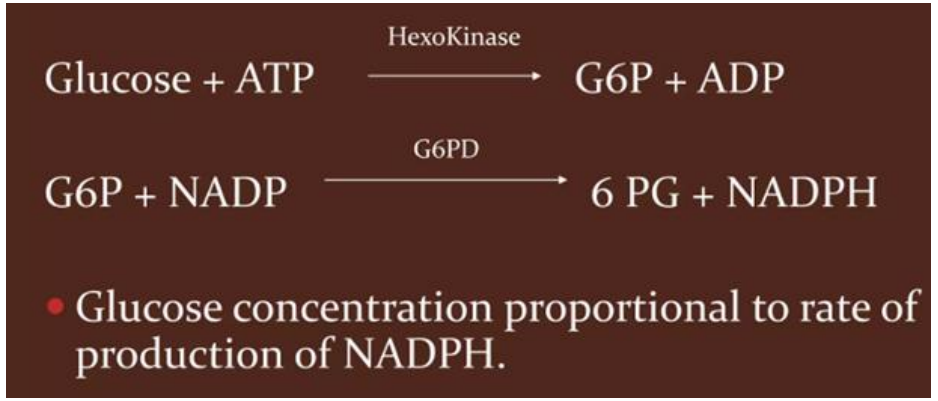
- Folin wu
- Somogyi-nelson method
- Orthotoluidine method

Enzymatic method

- Glucose oxidase/peroxidase
- **Hexokinase**

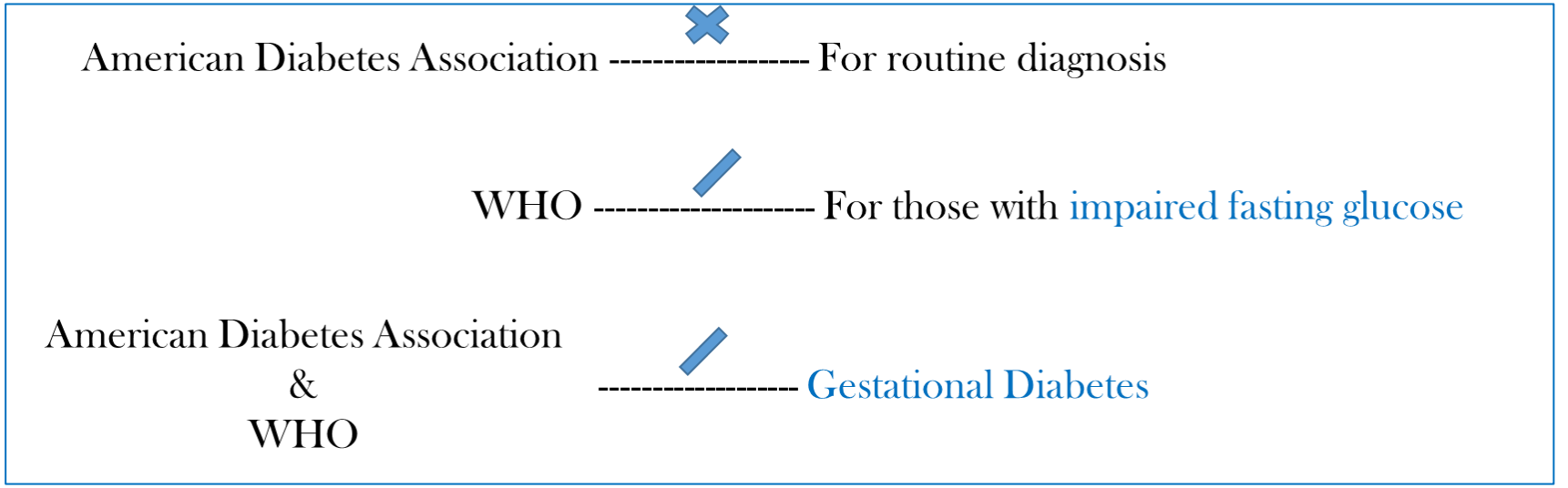
Advantages of enzymatic method

- Accurate, sensitive, specific and precise
- Reagents are safe to handle
- Very small plasma & reagents are required
- Linear up to 700 mg/dL
- NaF do not interfere in the assay



Oral Glucose Tolerance Test (OGTT)

Glucose tolerance means the ability of body to utilize glucose in blood circulation.



Patient Preparation

- Patient should on carbohydrate rich unrestricted diet for 3 days
- Patient should be ambulatory with normal physical activity
- Medications should be discontinued on the day of testing
- Exercise, smoking, tea and coffee are not allowed during test period.
- OGTT carried out in the morning after patient has fasted overnight for ~8 hours

OGTT: For patient with *Impaired Fasting Glucose*

➢ 75-g OGTT

Procedure

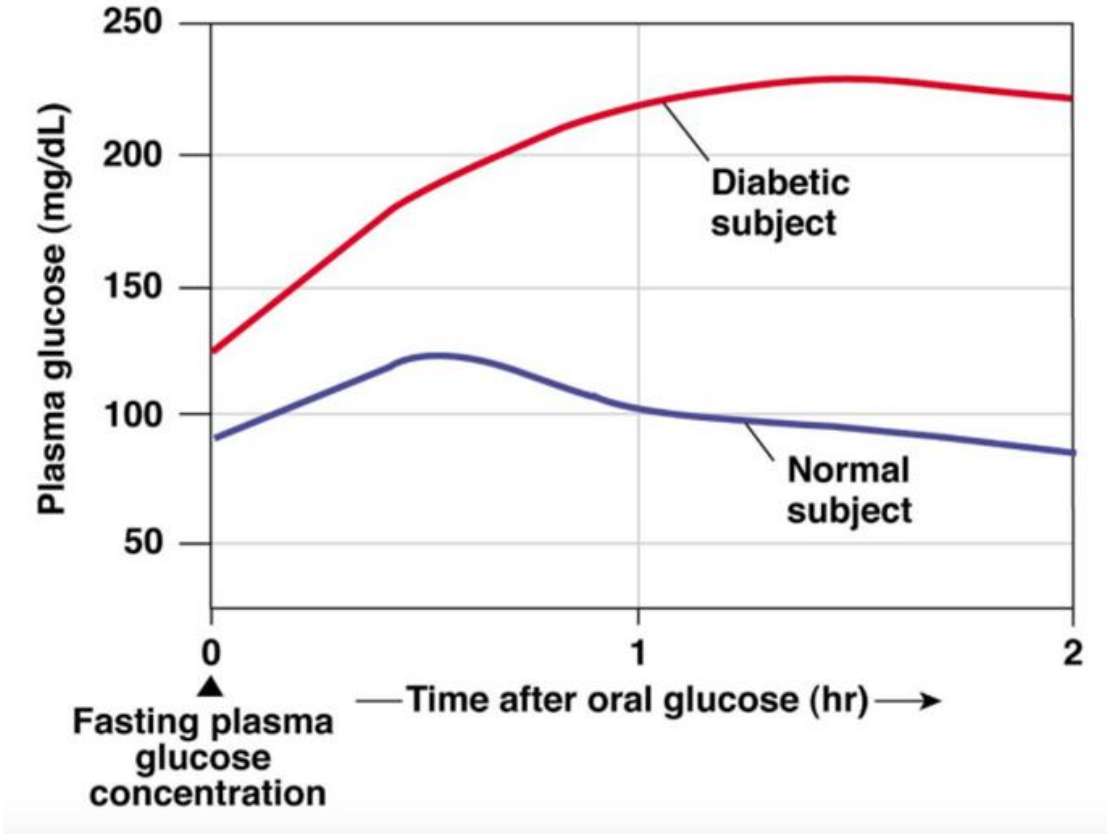
1. A fasting venous blood sample is collected in the morning.
2. Patient ingests 75 g of anhydrous glucose in 250-300 mL of water over 5 minutes.

   (For children, the dose is 1.75 g of glucose per kg of body weight up to maximum 75 g of glucose).

   Time of starting glucose drink is taken as 0 hour.
3. A single venous blood sample is collected 2 hours after the glucose load.
4. Plasma glucose is estimated in fasting and 2-hour venous blood samples.

OGTT: For patient with *Impaired Fasting Glucose*

➢ **75-g OGTT**



Normal	Fasting: <100 mg/dL 2-h OGTT: <140 mg/dL
Impaired fasting glucose	Fasting: 100-125 mg/dL 2-h OGTT: <140 mg/dL
Impaired glucose tolerance	2-h OGTT: 140-199 mg/dL
DM	Fasting: ≥ 126 mg/dL 2-h OGTT: ≥ 200 mg/dL

Factors affecting OGTT

- ❑ Acute infections
- ❑ ill patients
- ❑ Hypothyroidism
- ❑ Starvation

Gestational Diabetes Mellitus (GDM)



Impairment of glucose tolerance develops normally during pregnancy, particularly in 2nd and 3rd trimester.

ADA guidelines for laboratory diagnosis of GDM

Low risk	No need to be tested if *all* of the following criteria are met [Age <25 yrs, normal body weight (before pregnancy), absence of diabetes in first-degree relatives, member of an ethnic group with low prevalence of DM, no history of poor obstetric outcome, and no history of abnormal glucose tolerance]
Average risk	▪ Average-risk pregnant women are in between low and high risks ▪ Tested at 24-28 weeks of gestation
High risk	Who meet *any one* of the following criteria should be tested immediately [Marked obesity, strong family history of DM, glycosuria, or personal history of GDM]

Gestational Diabetes Mellitus (GDM)

Initial with fasting plasma glucose (FPG)

- FPG ≥ 126 mg/dL → repeat FPG testing on subsequent day for confirmation of DM
- Normal FPG → Average-risk and high-risk pregnant women → OGTT

OGTT: For patient with *GDM*

There are two approaches for laboratory diagnosis of GDM.

One-step approach	• A fasting patient undergoes a 2-hour, 75-g oral glucose tolerance test • The result is "abnormal" if any one glucose result (the fasting, 1-hour, or 2-hour result) is above a specified threshold.
Two-step approach	• Screening with 50g glucose challenge test + Diagnosis with 100g OGTT

Two-step Approach for Diagnosis of *GDM*

1. Screening

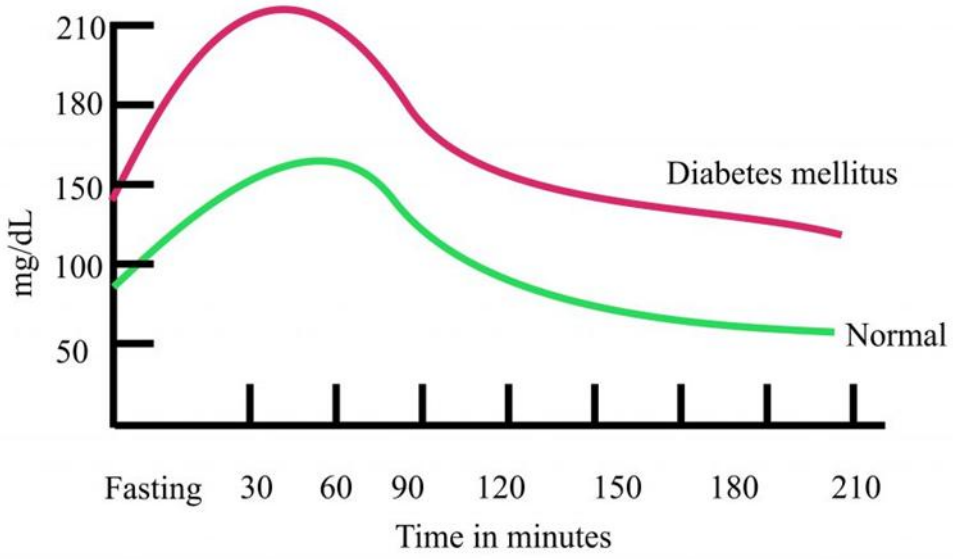
➢ **Glucose challenge test (50-g GCT)**

- A 50g glucose drink & collect blood sample at 1 hour later
- Glucose level <140 mg/dL → exclude GDM
- Glucose level ≥140 mg/dL → 100g OGTT

2. Diagnosis

➢ **100-g OGTT**

- Collect blood sample in the morning after 8 h fasting (FPG) and after drinking 100g glucose at 1, 2 and 3 hours



เวลา	National Diabetic Data Group (NDDG) (≥ 2 abnormal results = GDM)	Carpenter & Coustan (≥ 2 abnormal results = GDM)
Fasting	105 mg/dL	95 mg/dL
1-hour	190 mg/dL	180 mg/dL
2-hour	165 mg/dL	155 mg/dL
3-hour	145 mg/dL	140 mg/dL

Diagnosis of GDM: abnormal ≥ 2 values

Laboratory Test for Screening of DM

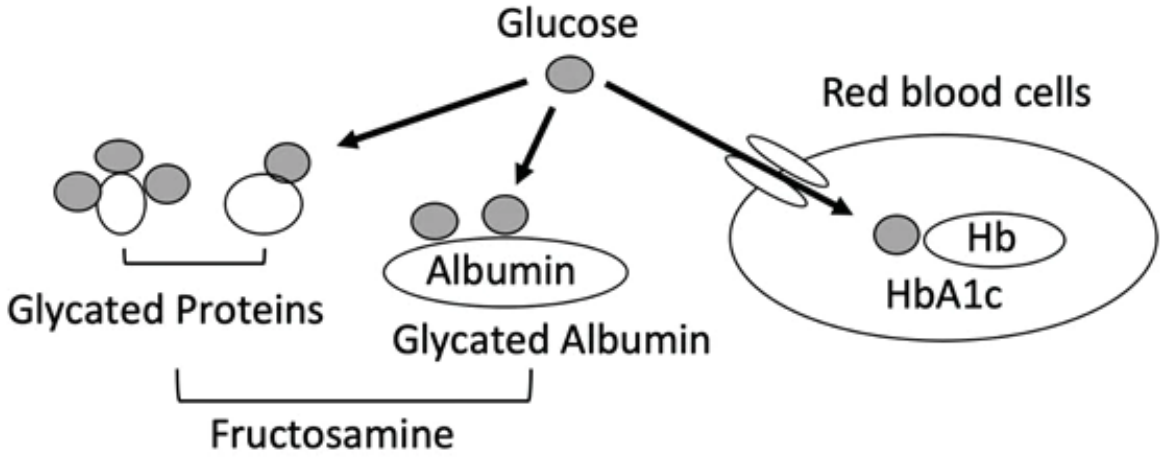
- Aim of screening is to identify asymptomatic individuals who are likely to have DM.
- Since early detection and prompt institution of treatment can reduce subsequent complications of DM, screening may be an appropriate step in some situations.

1. ผู้ที่มีอายุ 35 ปี ขึ้นไป
2. ผู้ที่อ้วน (BMI ≥ 25 kg/m^2 และ/หรือ มีรอบเอวเกินมาตรฐาน) และญาติสายตรง (พ่อ แม่ พี่ หรือน้อง) เป็นโรคเบาหวาน
3. ความดันโลหิตสูง (≥ 140 90 mmHg)
4. มีระดับไขมันในเลือดผิดปกติ (HDL cholesterol ≤ 35/mg dL และ/หรือ ไตรกลีเซอไรด์ ≥ 200 mg/dL) หรือได้รับยาลดไขมันอยู่
5. สตรีที่มีประวัติเป็นโรคเบาหวานขณะตั้งครรภ์ หรือเคยคลอดบุตรที่มีน้ำหนักตัวแรกคลอด เกิน 4 กิโลกรัม
6. มีโรคหัวใจและหลอดเลือด (Cardiovascular Disease)
7. สตรีที่มีภาวะถุงน้ำรังไข่ (Polycystic Ovary Syndrome)
8. เคยตรวจพบว่ามีภาวะ impaired glucose tolerance (IGT) หรือ impaired fasting glucose (IFG)

- Recommended screening test is fasting plasma glucose (FPG).
- If FPG is normal, screening test should be repeated every 3 years.

Laboratory Tests for Assessment of Glycemic Control

- Measurements of Glycated Proteins → Long-term glycemic control
  - Hemoglobin A1c (HbA1c)
  - Fructosamine



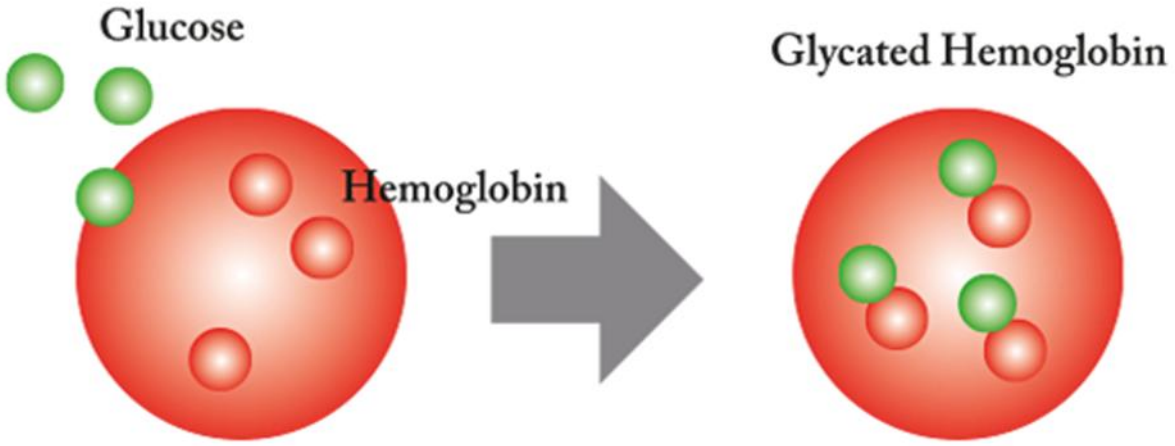
- Daily Self-Monitoring of Blood Glucose (SMBG) → Day-to-day or immediate control



Laboratory Tests for Assessment of Glycemic Control

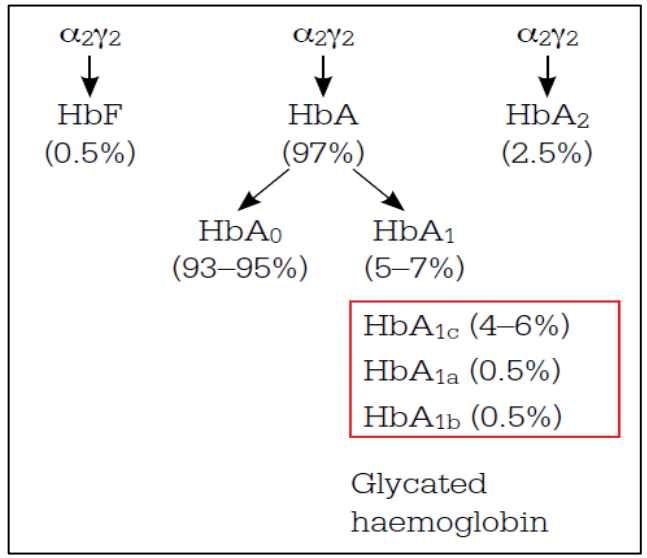
		HbA 1c	Glycated Albumin	Fructosamine
Analyte	Stucture	Glycated valine of N-terminal the β-chain of hemoglobin	Glycated lysine of Albumin	Fructosamine of serum proteins
	Location	Red Blood Cell	Whole Body	Whole Body
Unit		%(mmol/mol)	%(mmol/mol)	μmol/L
Duration of glycemia reflected		Long (2-3 months)	Intermediate (2-3 weeks)	Intermediate (2-3 weeks)
Strengths		*Gold Standard for monitoring and diagnose of diabetes. *Low within-person variablinty	*Monitoring the effects of changes in therapy *Pregnancy *Dialysis *Hemoglobinopethies, *Iron deficiency *Anemia *Low within-person variability	*Monitoring the effects of changes in therapy *Pregnancy *Dialysis *Hemoglobinopethies, *Iron deficiency *Anemia
Limitations		*Short term monitoring *Hemoglobinopathies *Iron deficiency *Anemia *Hemodialysis *Pregnancy	*Nephrosis *Liver cirrhosis *Thyroid disease	*Has not been standardized *Little evidence *Affected by serum protein concentration and proportion *Nephrosis *Liver cirrhosis *Thyroid disease

Hemoglobin A1c (HbA1c)



- A red blood cell (RBC) lives for approximately 120 days
- Hemoglobin (Hb) is contained in the RBC.
- Glucose (sugar) enters in the bloodstream and the RBC.

- Glucose naturally binds to Hb.
- This binding creates glycated Hb (HbA1c)



Type	Components
HbA1a1	Fructose-1, 6-biphosphate bound to the N-terminal valine of the β-chain
HbA1a2	Glucose-6-phosphate bound to the N-terminal valine of the β-chain
HbA1b	Pyruvic acid bound to the N-terminal valine of the β-chain
HbA1c	Glucose bound to the N-terminal valine of the β-chain

Hemoglobin A1c (HbA1c)

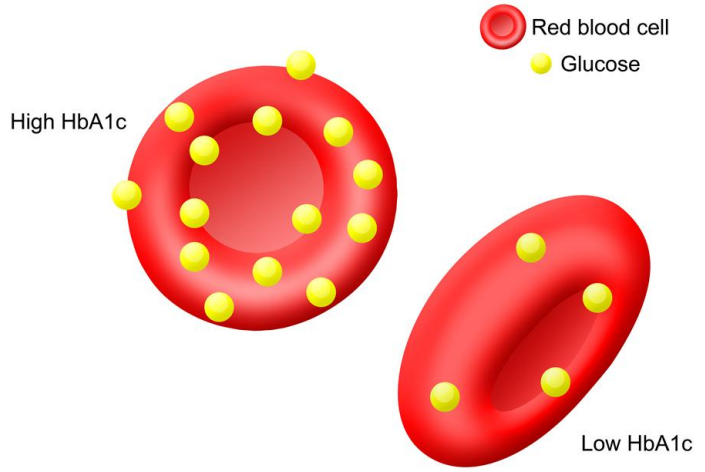
Concentration of HbA1c in blood

depend on:

- RBC life span



- Blood glucose concentration



With every rise of 1.0% HbA1c, plasma glucose increases by 35 mg/dL

HbA1c in DM

- In all diabetics to monitor long-term blood glucose level control

  >> If HbA1c values >8%, the treatment should be reconsidered.



- To monitor patient compliance

	HbA1c Test Score	Mean Blood mg/dL	Glucose mmol/L
Excellent	4.0	50	2.6
	5.0	80	4.7
	6.0	115	6.3
Good	7.0	150	8.2
	8.0	180	10.0
Poor	9.0	215	11.9
	10.0	250	13.7
	11.0	280	15.6
	12.0	315	17.4
	13.0	350	19.3
	14.0	380	21.1

- To predict development & progression of microvascular complications
- For determining the therapeutic option: oral agents, insulin, β-cell transplantation
- HbA1c testing allows for more timely treatment changes

Frequency of HbA1c testing

- At least 2 times each year: in individuals meeting treatment targets and have stable glycemic control
- Quarterly: in individuals whose therapy has changed or who are not meeting glycemic targets

Measurement of HbA1c



EDTA whole blood

- No effect of diet, exercise & insulin on test results
- Fasting is not required
- Blood sample can be drawn at any time of day

➢ HbA1c is reported as

NGSP

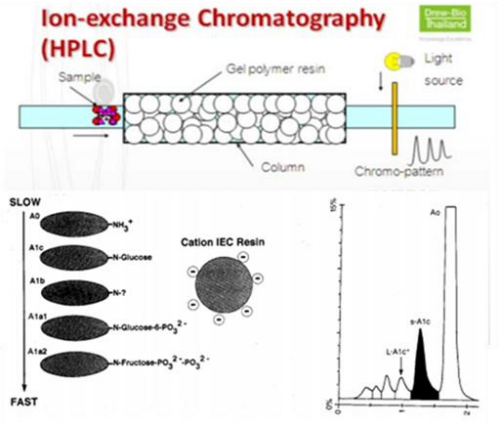
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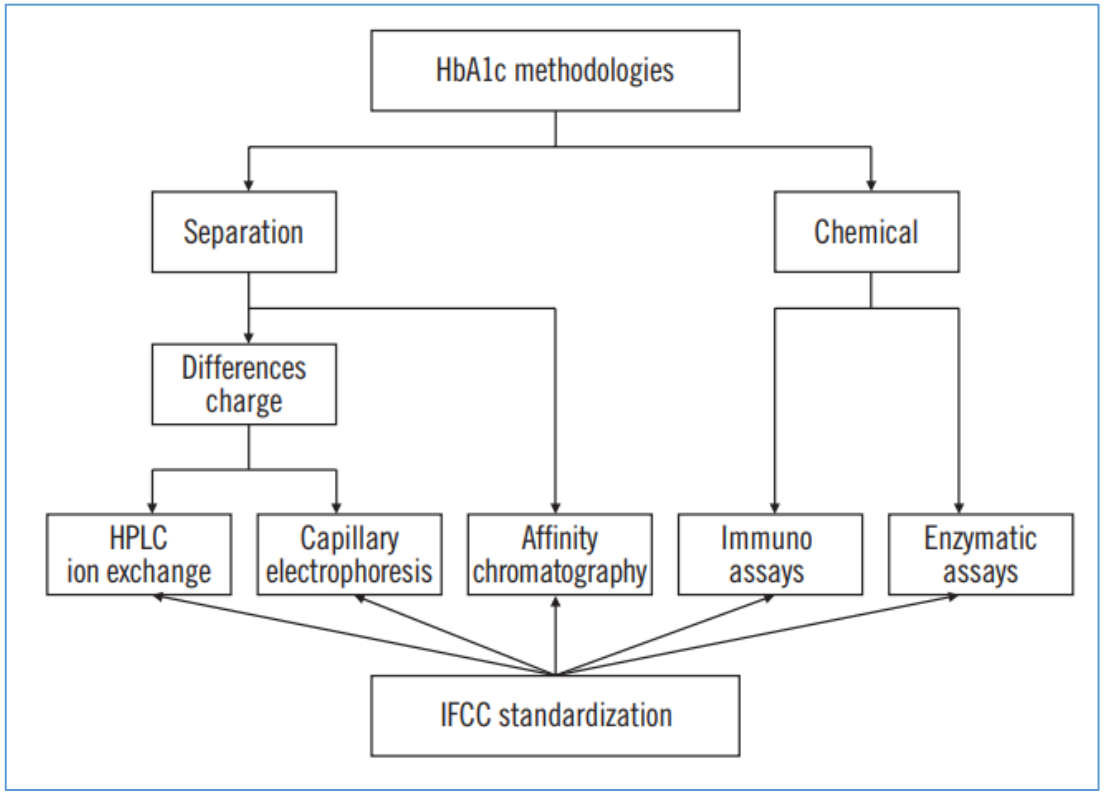
IFCC International Federation of Clinical Chemistry and Laboratory Medicine

mmol/mol

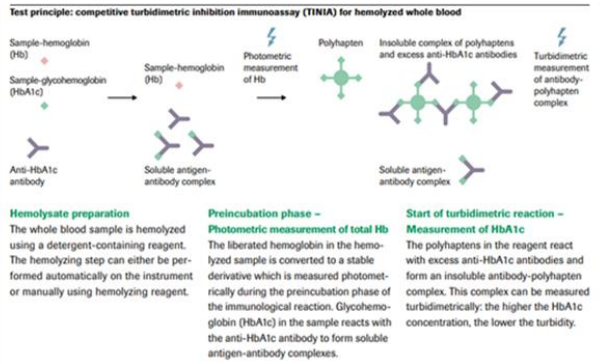
Methods for HbA1c Measurement

HPLC

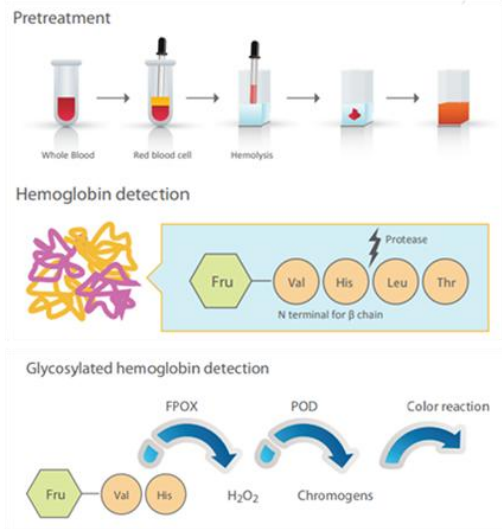




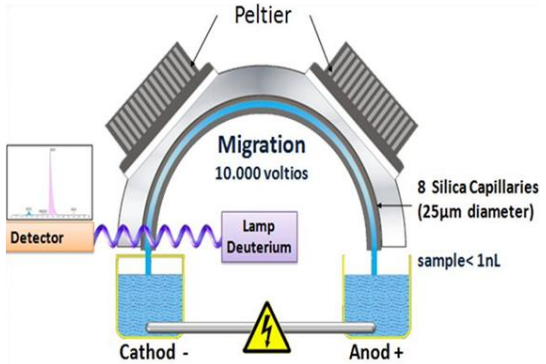
Immunoassay

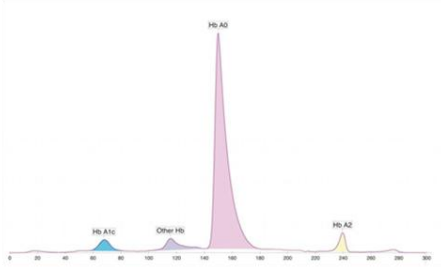


Enzymatic assay



Capillary electrophoresis (CE)





Factors Interfering HbA1c Results

Disease or condition	Effect on HbA_{1c} level	Potential mechanism
Rapid erythrocyte turnover	Falsely low	Unstable erythrocyte pool
Hemolytic states	Falsely low	Unstable erythrocyte pool
Iron deficiency anemia	Falsely high	Unknown
Hemoglobin SS, SC or CC disease	Falsely low	Unstable erythrocyte pool
Variant hemoglobin trait	Variable	Assay interference
Fetal hemoglobin	Variable	Assay interference
Blood transfusions	Falsely low	Unstable erythrocyte pool
Aging	Falsely high	Unknown
Cirrhosis	Falsely low	Unknown
Uremia	Falsely low	Carbamylated hemoglobin
Hemodialysis	Falsely low	Multiple
HIV infection	Falsely low	Occult hemolysis
Pregnancy	Falsely low	Hemodilution?
Dyslipidemia	Variable	Assay interference
Hyperbilirubinemia	Variable	Assay interference
Aspirin use (large doses)	Variable	Acetylated hemoglobin
Vitamin C	Variable	Interference with glycation
Vitamin E	Falsely low	Interference with glycation
Alcohol excess	Variable	Assay interference
Opiate use	Variable	Assay interference

Factors Interfering HbA1c Results

- Factors that interfere with HbA1c testing:
  e.g. HbF, Hb variants

- Factors that affect the interpretation of HbA1c results



age, race/ethnicity









IDA: malonidaldehyde ↑ → glycation ↑

Hb variant: substitution at a glycation site

Decreased	Increased
▪ Iron supplementation ▪ Hemolytic anemia ▪ Pregnancy (2nd, 3rd trimester) ▪ G6PD deficiency ▪ Hemoglobinopathies ▪ Rheumatoid arthritis ▪ Splenomegaly ▪ Hemodialysis ▪ Recent blood loss ▪ Blood transfusion ▪ Erythropoietin therapy	▪ Iron deficiency ▪ B12 or folate deficiency ▪ Splenectomy

Factors Interfering HbA1c Results

Conditions to suspect that there is an interference of HbA1c

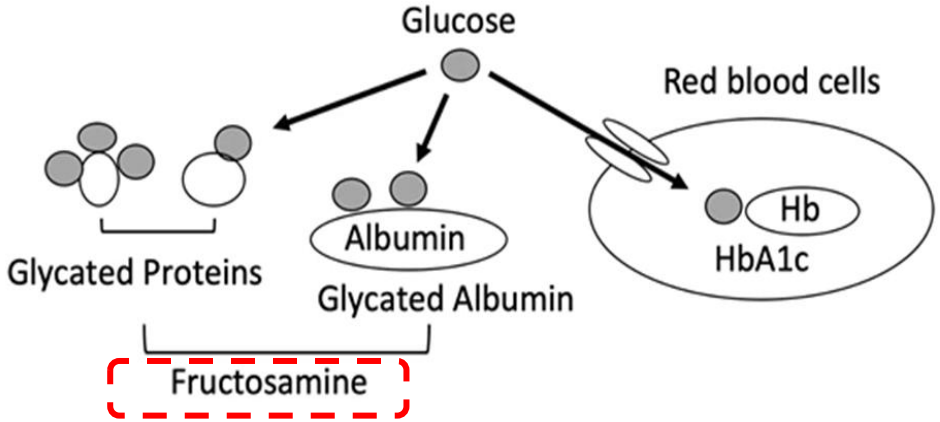
- Notable value: below the lower limit of reference range or higher the upper measuring limit
- Abnormal finding: chromatogram or electropherogram
- Inconsistent match: clinical features, other test results

HbA1c cannot be measured in individuals who do not have HbA, *e.g.*, those with homozygous Hb variants, such as HbSS, HbEE

What to do?

Alternative measure e.g.
- Glucose monitoring
- Glycated protein (fructosamine) or glycated albumin

Fructosamine



		Fructosamine
Analyte	Stucture	Fructosamine of serum proteins
	Location	Whole Body
Unit		µmol/L
Duration of glycemia reflected		Intermediate (2-3 weeks)
Strengths		*Monitoring the effects of changes in therapy *Pregnancy *Dialysis *Hemoglobinopethies, *Iron deficiency *Anemia
Limitations		*Has not been standardized *Little evidence *Affected by serum protein concentration and proportion *Nephrosis *Liver cirrhosis *Thyroid disease

- Measure average blood glucose level over the past 2-3 weeks

	ค่าปกติ	ค่าที่ใช้ในการติดตามการรักษา		
		ดี	พอใช้	ต้องปรับปรุง
Fructosamine (mg/dL)	< 258	259 – 279	280 – 320	> 320

Self-monitoring of Blood Glucose (SMBG)



SMBG devices measure capillary whole blood glucose obtained by finger prick and use test strips that used enzymatic methods or biosensor.

In some strips, a layer is incorporated to exclude blood cells so that glucose in plasma is measured.

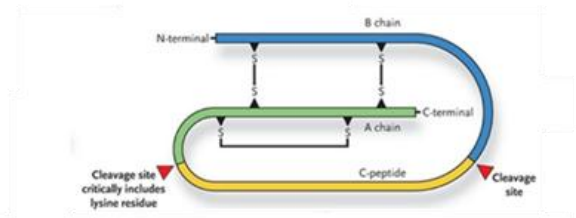
"Blood glucose meter; BGM"

Daily self assessment of blood glucose

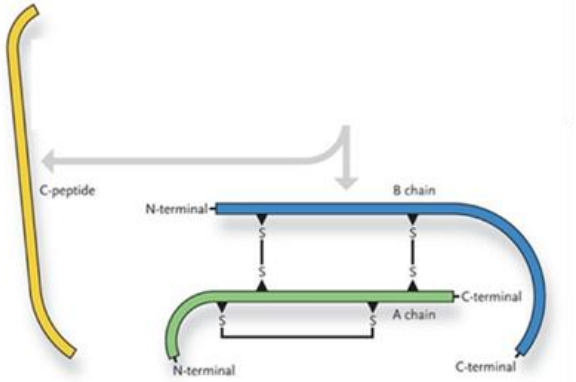
- Regular use of SMBG devices by diabetic patients has improved the management of DM.
- Daily use of SMBG devices can avoid major hypoglycemic episodes.

- SMBG devices yield unreliable results at very high and very low glucose levels.
- It is necessary to periodically check the performance of the glucometer by measuring parallel venous plasma glucose in the laboratory.

Other Tests



Proinsulin



Insulin

❑ Insulin assay

❑ Proinsulin assay

The biological activity of proinsulin is only about 10% of insulin, but the half life of proinsulin is three times as long as insulin.

❑ C-peptide assay

- Produced in equal amounts to insulin (1:1 ratio)
- **Longer haft-life (20-30 min)** than insulin (3-5 min)
- **5 to 10 times higher concentration** than insulin in the systemic circulation
- Not influenced by exogenous insulin

Best measurement of endogenous insulin secretion

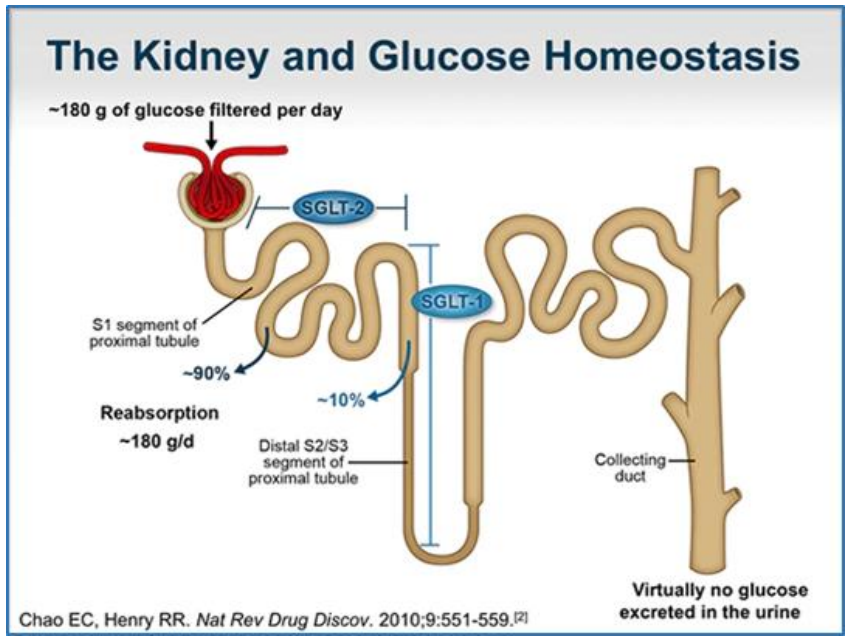
✓ To determine the cause of hypoglycemia >> Overdose of insulin or Insulinoma

✓ Classification of diabetes mellitus

✓ Therapeutic & Prognosis

Other Tests

❑ Estimation of Urine Glucose



Semiquantitative urine glucose testing for monitoring of diabetes mellitus in home setting is not recommended.

Presence of glucose in urine → Glycosuria

- ➢ Semi-quantitative test: negative, 1+, 2+, 3+, 4+
- ➢ Quantitative test



- Urine glucose test results correlate well with plasma glucose values.
- Presence of glucose in urine indicated that blood glucose level of the patient could have elevated >180 mg/dL

Renal glucosuria !!!
Other causes of glucosuria !!!

Laboratory Tests for Assessment of Long-term Risks

➢ Urinary albumin excretion or Microalbuminuria

Normally, only a small amount of albumin is filtered at the glomerulus, and most of that albumin is degraded and reabsorbed by the proximal tubule.

- It indicates increase in capillary permeability to albumin.
- Albumin: enter the urine after the kidney is damaged

➢ Early detection of diabetic nephropathy is based on estimation of urinary albumin excretion.

- **Diabetic nephropathy** develops in around 20-30% of patients with type 1 or type 2 DM.

❑ In all adult patients with DM, usual reagent strip test for proteinuria should be carried out periodically.

✓ Positive test means presence of overt proteinuria or clinical proteinuria and may be indicative of overt nephropathy. → quantitation of albuminuria to plan appropriate therapy

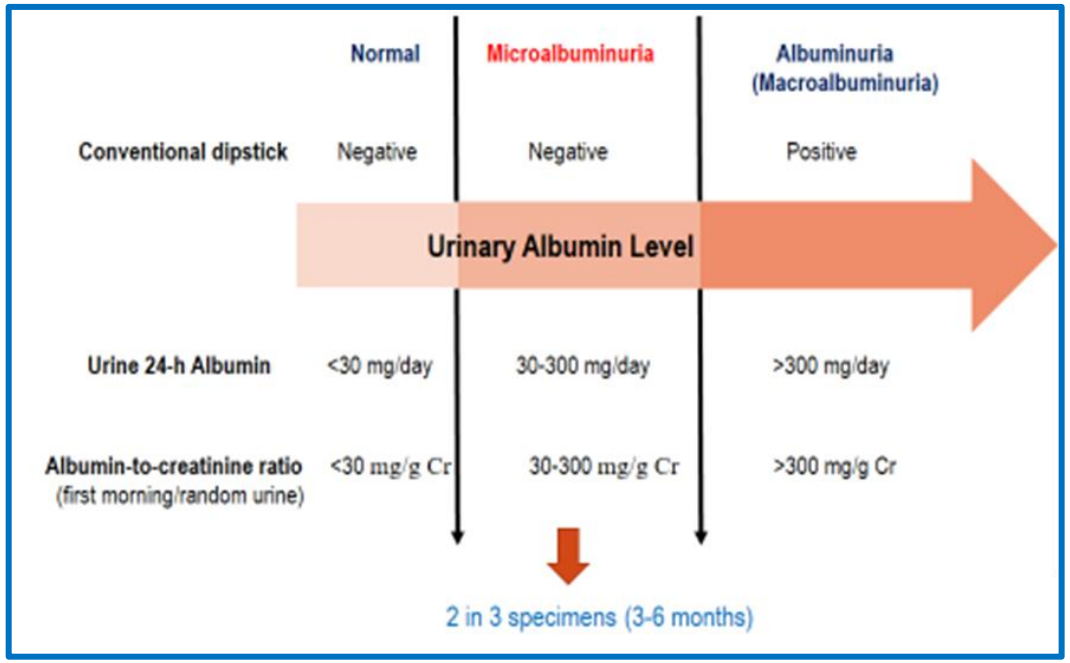
❖ If the routine dipstick test for proteinuria is negative → microalbuminuria test

Detection and Interpretation of Microalbuminuria (Urine albumin)

Measurements of microalbuminuria:

- Albumin to creatinine ratio in a random urine sample
- Urinary albumin excretion in an early morning and random sample
- Urinary albumin excretion in a 24-hour urine sample

Interpretation:



- It is the earliest marker of diabetic nephropathy. Early diabetic nephropathy is reversible.
- Appearance of microalbuminuria is predictor of progression to overt proteinuria.
- It is a risk factor for cardiovascular disease in both type 1 and type 2 patients.
- It is associated with higher blood pressure and poor glycemic control.

Laboratory Tests for Assessment of Acute Complications

The three most serious acute metabolic complications of DM are:

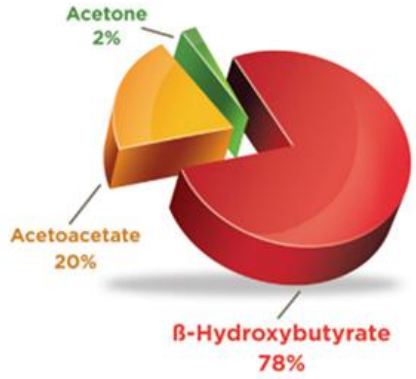
- Diabetic ketoacidosis (DKA)
- Hyperglycemic hyperosmolar state
- Diabetic hypoglycemia

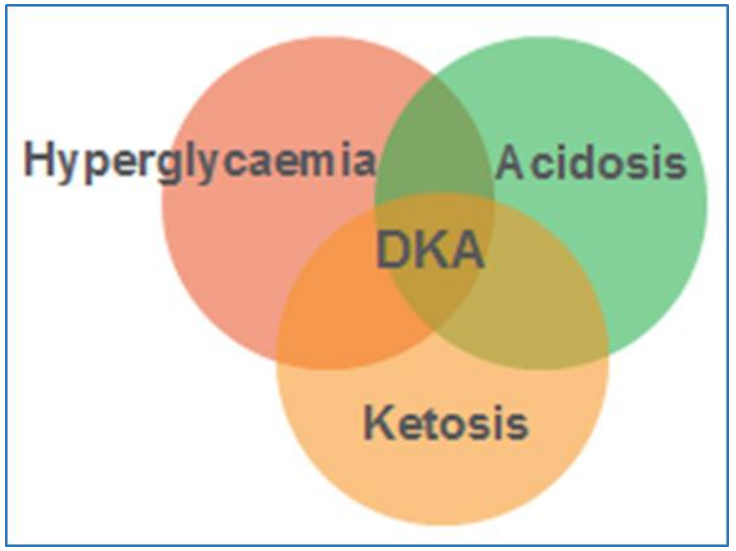
Diabetic Ketoacidosis (DKA)

- State of absolute or relative insulin deficiency aggravated by ensuring hyperglycemia, dehydration, and acidosis-producing derangements in intermediary metabolism.
- The common causes of DKA are infection, noncompliance with insulin therapy, alcohol abuse, and myocardial infarction
- More common in case of Type 1 DM
- Increased synthesis causes the accumulation of ketone bodies in blood.



➢ Develops over hours to 1-2 days

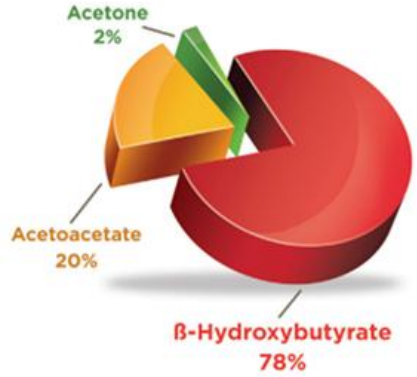




Lab findings in DKA

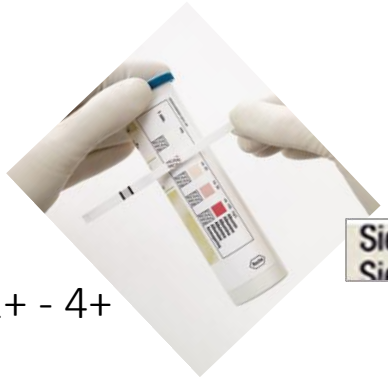
- Severe hyperglycemia: plasma glucose 300-500 mg/dL
- Increased blood/urine ketones
- pH <7.3
- Low bicarbonate
- High anion gap

Methods to detect Ketone Bodies

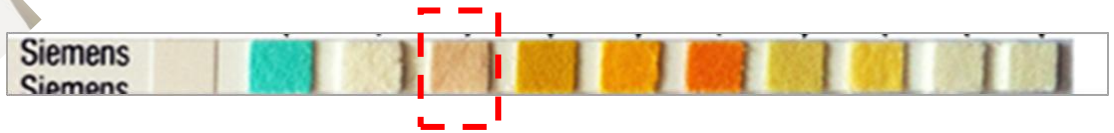


- Ketone bodies are formed from metabolism of free fatty acids.

➢ Reagent strip



- Specimens: Plasma & Urine
- Semi-quantitative: negative, 1+ - 4+



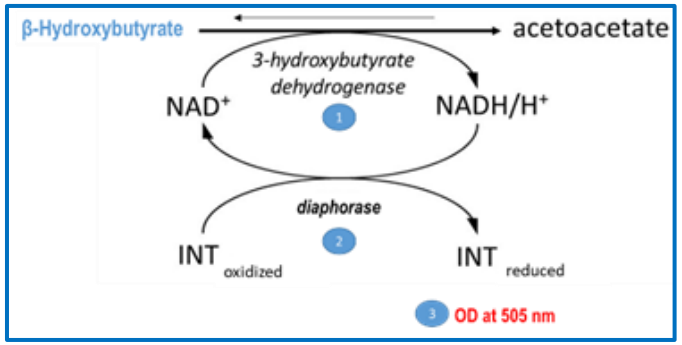
Use nitroprusside → measure only acetoacetate

- Test for urine ketones alone should not be used for monitoring of DKA.

➢ Colorimetric enzymatic method

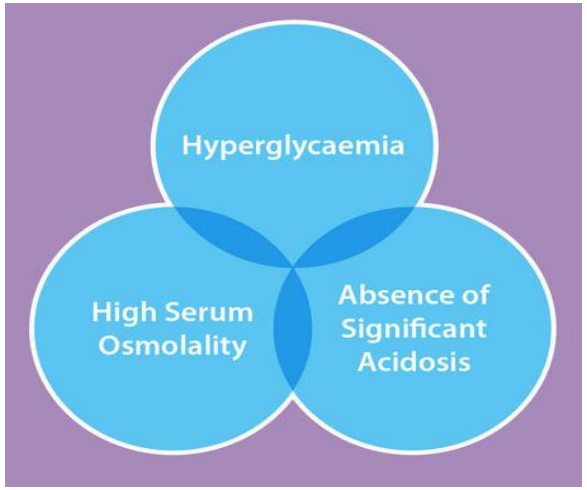
- Specimens: Plasma
- Quantitative test
- Measure β-hydroxybutyric acid

>> Use to diagnosis and monitoring DKA



Hyperglycemic Hyperosmolar State (HHS)

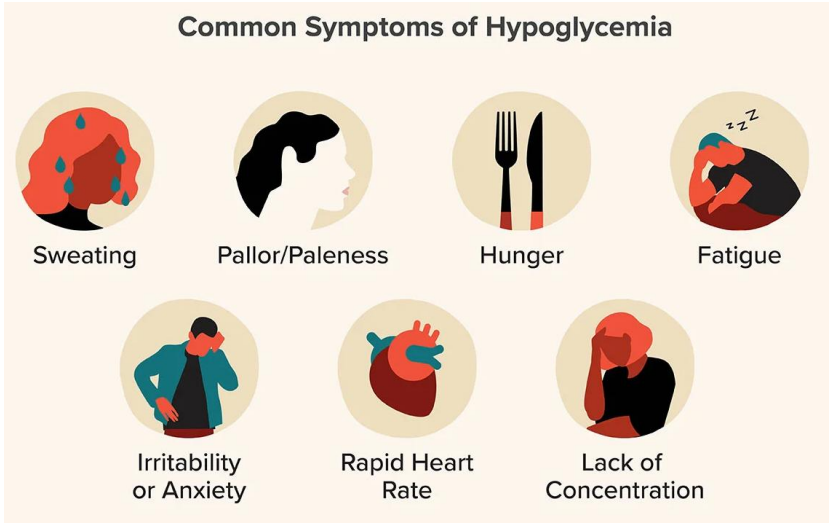
- Hyperosmolar hyperglycemic state (HHS) is characterized by: very high blood glucose level (> 600 mg/dL), hyperosmolality (>320 mOsmol/kg of water), dehydration, lack of ketoacidosis, and altered mental status.



- Insulin secretion is adequate to prevent ketosis but not hyperglycemia.
- Causes of HHS are illness, dehydration, surgery, and glucocorticoid therapy.
- More common in case of Type 2 DM
- Develops over days to weeks

Diabetic Hypoglycemia

- Results from an imbalance between glucose utilization and production in such a manner that the rate of glucose utilization exceeds the rate at which glucose being produced.



Causes:



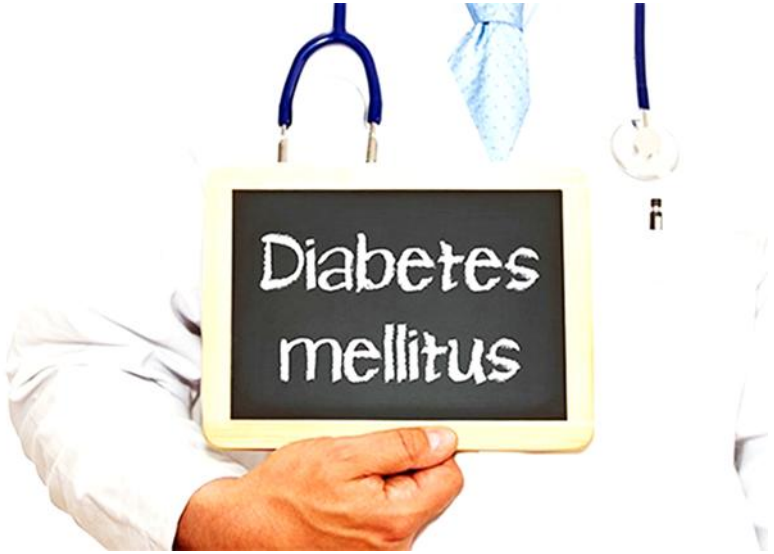
Whipple's triad

(1) **Symptoms** consistent with hypoglycemia

(2) **Low plasma glucose** measured with precise method
(DM: < 70 mg/dL, Non-DM: < 50 mg/dL)

(3) Relief of those symptoms if plasma glucose is raised

Conclusion



Diabetes is a very complicated disease.

Laboratory plays an important part in the diagnosis and management of diabetic patient.

Laboratory Tests for DM

- Diagnosis of DM
- Screening of DM
- Assessment of glycemic control
- Assessment of long-term risks
- Assessment of acute metabolic complications

Blood

- Plasma glucose: random, fasting, 2-h postprandial
- Glucose tolerance test (GTT)
- HbA_1c
- Fructosamine
- C-peptide

etc.

Urine

- Glucose
- Microalbumin
- Ketone

etc.

POST-TEST EXAMINATION





sriwimol.wi@gmail.com


Thank you
for your
attention